

Integrating using inverse trig

Key Points

By considering the derivatives of inverses we get integrations to do with the reciprocal of squares.

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1}\left(\frac{x}{a}\right) = -\cos^{-1}\left(\frac{x}{a}\right)$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1}\left(\frac{x}{a}\right)$$

Basic Skills

Q1. Find the derivatives of

a) $\arctan x$

b) $\arcsin x$

c) $\arccos x$

Q2. Explain why we don't really use the differential of $\cos^{-1} x$ for integrating difference of squares

Q3. Using an appropriate integration formula find

a) $\int \frac{1}{25+x^2} dx$

b) $\int \frac{1}{\sqrt{16-x^2}} dx$

Competency Skills

Q4. Evaluate the following integrals

a) $\int \frac{1}{125+5x^2} dx$

b) $\int \frac{1}{\sqrt{40-2x^2}} dx$

Q5. Evaluate the following giving your answer in logarithm form

a) $\int_0^1 \frac{1}{\sqrt{9-3x^2}} dx$

b) $\int_1^{\sqrt{3}} \frac{5}{3+x^2} dx$

c) $\int_0^{\frac{1}{2}} \frac{x}{-2x^4-60} dx$

Q6. Find the exact area under the graph $y = \frac{1}{\sqrt{25-(2x+4)^2}}$ between $x = -\frac{9}{2}$ and $x = \frac{1}{2}$

Q7. Find the area between the curves $y = \frac{3}{x^2+1}$ and $y = \frac{1}{\sqrt{2-x^2}}$

Advanced Skills

Q8. Evaluate the integrals

a) $\int_{-1}^2 \frac{1}{\sqrt{4x+5-x^2}} dx$

b) $\int \frac{1}{20-6x+2x^2} dx$

Q9. Prove $\int \frac{1}{\sqrt{1-x^2}} dx = \sin^{-1}(x)$ using the substitution that $x = \sin u$

Q10. Prove $\int \frac{1}{1+x^2} dx = \tan^{-1}(x)$ using the substitution that $x = \tan u$

Q11. Find the total area bound by the curve $y = \frac{1}{1+x^2}$ and the x -axis

Q12. Find $\int \sqrt{1-x^2} dx$ by using an appropriate substitution

Extension

Q13. Explain why standard inverse trig functions cannot be used to find the integral $\int \frac{1}{\sqrt{x^2-a^2}} dx$